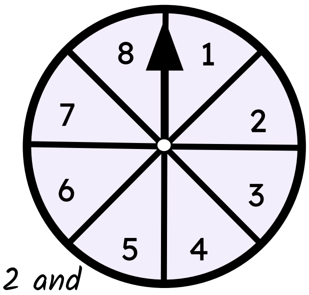
Summing probabilities

Task A: Summing probabilities of events

1) The spinner is spun once. For each pair of events listed below, state whether their probabilities would sum to 1. Justify your answers with reasoning.



a) A = {odd}, B = {even}

*Probabilities sum to 1 because the events are **mutually exclusive** and **exhaustive**.*

b) A = {odd}, B = {multiple of 4}

*Probabilities do not sum to 1 because the events are not **exhaustive** (e.g. 2 and 6 are neither odd or a multiple of 4).*

c) A = {odd}, B = {factor of 24}

*Probabilities do not sum to 1 because the events are not **mutually exclusive** (e.g. 1 and 3 are both odd and a factor of 24).*

d) A = {odd}, B = {prime}

*Probabilities do not sum to 1 because the events are not **mutually exclusive** (e.g. 3, 5 and 7 are both odd and prime).*

*OR Probabilities do not sum to 1 because because the events are not **exhaustive** (e.g. 4, 6 and 8 are neither odd or prime).*

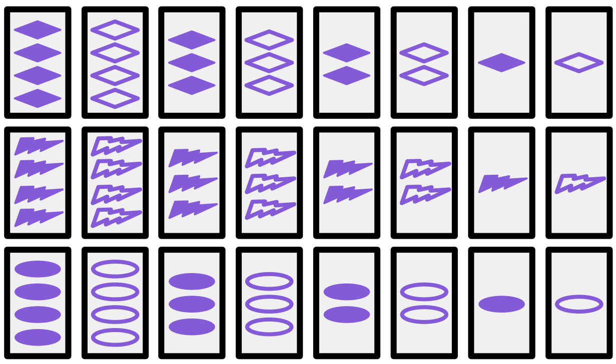
e) A = {integer}, B = {non-integer}

*Probabilities sum to 1 because the events are **mutually exclusive** and **exhaustive**.*

2) A game involves the set of cards below. Each card shows one of three shapes.

The three shapes: rhombus, lightning, oval

Shapes are either solid or hollow.
E.g. solid hollow



From the list of events below, find sets of events where the probabilities sum to 1.

A = {rhombus} B = {1 shape} C = {solid} D = {lightning}

E = {2 shapes} F = {hollow} G = {3 shapes} H = {oval}

I = {4 shapes} J = {not oval} K = {more than 1 shape}

A, D and H {rhombus, lightning, oval}

H and J {oval, not oval}

B, E, G and I {1 shape, 2 shapes, 3 shapes, 4 shapes}

B and K {1 shape, more than 1 shape}

C and F {solid, hollow}

Task B: Finding missing probabilities

1) Find the value of each unknown probability.
Each question presents a complete set of possible outcomes for a trial.

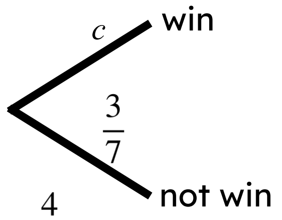
a) P(rain) = 25%
P(not rain) = a

a = 75%

b) P(train on time) = 0.6
P(train late) = b

b = 0.4

c)

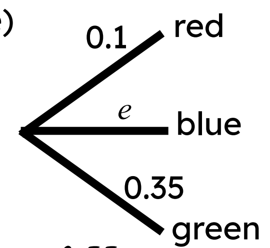


$c = \frac{4}{7}$

d) P(Player 1 wins) = 24%
P(Player 2 wins) = 32%
P(draw) = d

d = 44%

e)



e = 0.55

f)

outcome	A	B	C	D
probability	$\frac{1}{6}$	$\frac{1}{4}$	f	$\frac{3}{12}$

$f = \frac{1}{3}$

2) Izzy, Laura and Sofia compete in a swimming race.
The outcomes and probabilities are represented in the table.

winner	Laura	Andeep	Sofia
probability	x	$2x$	0.4

Find the value of x.

$$x + 2x + 0.4 = 1$$
$$3x + 0.4 = 1$$
$$3x = 0.6$$
$$x = 0.2$$

3) Aisha, Jun, Lucas and Sam compete against each other on a video game.

Jun is as likely to win as Aisha.
Lucas is twice as likely to win as Jun.
Sam is three times as likely to win as Lucas.

What is the probability that Sam wins?

P(Aisha) = x
P(Jun) = x
P(Lucas) = $2x$
P(Sam) = $6x$

$$x + x + 2x + 6x = 1$$
$$10x = 1$$
$$x = 0.1$$
$$6x = 0.6$$